CASE STUDY 13

ATM Transaction Sequence Simulation

1. PROBLEM STATEMENT

Simulate a series of deposits and withdrawals, maintain an account balance, reject invalid transactions, enforce a

withdrawal limit, and display a transaction summary.

2. OBJECTIVES

• Process deposit and withdrawal operations in sequence.

• Prevent negative or excessive withdrawals.

• Maintain the balance and count successful transactions.

3. ASSUMPTIONS

• Opening balance is ■5,000.

• Each transaction is represented by a tuple containing type and amount.

• Amounts must be positive.

• A single withdrawal cannot exceed ■3,000 or the available balance.

4. ALGORITHM

1. Set the opening balance to ■5,000 and initialize transaction counters.

2. Read the sequence of deposit and withdrawal transactions.

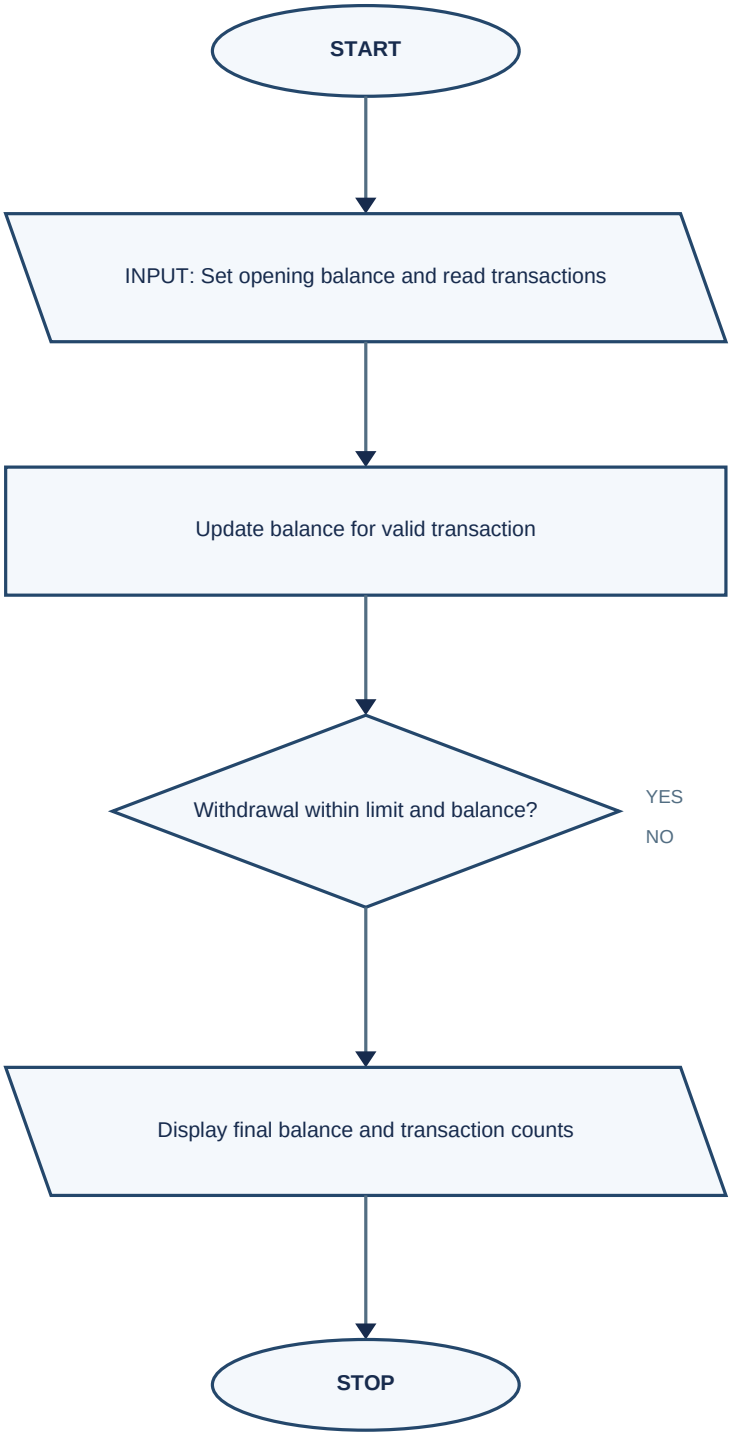
3. Reject transactions with non-positive amounts or unknown transaction types.

4. For withdrawals, check the ■3,000 limit and available balance before updating the balance.

5. Display the final balance and successful and rejected transaction counts.

5. FLOWCHART

Oval: START/STOP | Parallelogram: input/output | Rectangle: process/calculation | Diamond: decision



6. PSEUDOCODE

```
BEGIN
SET balance = 5000
SET successful = 0, rejected = 0
FOR each transaction (type, amount)
IF amount <= 0: reject transaction
ELSE IF type is deposit: add amount to balance
ELSE IF type is withdrawal
IF amount <= 3000 AND amount <= balance
subtract amount and accept
ELSE: reject transaction
ELSE: reject unknown transaction
DISPLAY final balance and transaction counts
END
```

7. PYTHON SOURCE CODE

```
transactions = [
("deposit", 2000),
("withdraw", 1500),
("withdraw", 4000),
("deposit", 500),
("withdraw", 1000)
]
balance = 5000
successful = 0
rejected = 0
for transaction in transactions:
kind, amount = transaction
if amount <= 0:
print("Rejected: invalid amount", amount)
rejected += 1
elif kind == "deposit":
balance = balance + amount
successful += 1
print("Deposit successful:", amount)
elif kind == "withdraw":
if amount <= 3000 and amount <= balance:
balance = balance - amount
successful += 1
print("Withdrawal successful:", amount)
else:
print("Withdrawal rejected:", amount)
rejected += 1
else:
print("Rejected: unknown transaction")
rejected += 1
print("Final balance: Rs.", balance)
print("Successful transactions:", successful)
print("Rejected transactions:", rejected)
```

8. TEST CASES & EXPECTED RESULTS

TEST INPUT / CONDITION REASONING EXPECTED RESULT

Deposit ■2000 Positive deposit Accepted; balance ■7000

Withdraw ■4000 Above single withdrawal limit Rejected

Withdraw ■1000 Within limit and balance Accepted

9. SAMPLE OUTPUT

```
Deposit successful: 2000
Withdrawal successful: 1500
Withdrawal rejected: 4000
Deposit successful: 500
Withdrawal successful: 1000
Final balance: Rs. 5000
Successful transactions: 4
Rejected transactions: 1
```

10. CONCLUSION

The ATM simulation updates the balance safely, applies transaction rules, and reports successful and rejected operations.

SYLLABUS CONCEPTS Variables • Data Types • Operators • Conditions • Loops • Lists • Nested Lists • Tuples • String Operations